

Agricultural Commodity Trading Platform

Trader Journey — Chronological Requirements & Explanations

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0) Context & Roles

Stakeholders and systems participating in the Trader flow after sign-in.

- Trader (account owner)
- Buyer (end customer)
- Commission Agent (lists on behalf of Producer)
- Producer (origin)
- Employees: Buyer/Procurement, Quality Verifier, Weighing Operator
- Grok AI: insights, visual verification, auto-bid, anomaly detection
- Security: OTP confirmations for binding steps; immutable audit logs

1) After Sign-In (Trader Landing)

Buyer Offers Inbox shows actionable buyer offers mapped to commodities/qualities/volumes.

- Buyer identity + rating + payment discipline history
- Commodity → Variety → Grade (e.g., Chilli → Teja → Export) and volume
- Price structure (target / fixed / $\pm$ band), packaging, weight-loss tolerances
- Transport basis (Delivered / Actuals / Buyer-arranged); Insurance basis
- Tax basis (inclusive/exclusive; jurisdiction auto-computed)
- Payment terms preset (dropdown) and regulatory due dates
- Primary actions: Accept (OTP), Counter (negotiate), Request Advance, Decline

Commodity-Aware Request Parameters

Packaging

- Bag types (Jute/PP/HDPE/Carton/Vacuum/Loose), sizes (25/40/50/75 kg)
- Cost basis: Included / Excluded / Buyer-provided
- Marks/labels: gross/net on bag, batch codes, QR/token, origin marks

Weight & Loss Tolerances

- Acceptable weight loss % (per bag or lot; commodity templates)
- Moisture tolerance (upper bound) and test method (digital/oven-dry)
- Impurity/foreign matter, broken, stems (commodity-specific fields)

Pricing, Transport & Insurance

- Basis: Ex-warehouse / FOB (loaded) / Delivered

- Transport: Included / Actuals extra / Buyer-arranged
- Insurance: Seller / Buyer / Included / None

Tax & Jurisdiction

- Price inclusive/exclusive of taxes (GST/VAT/local cess)
- IGST vs CGST+SGST computed by place of supply; reverse charge toggle
- Export flags: LUT/Bond and export documentation

Payment Terms (presets + custom)

- Advance %, balance timing; credit 7–90 days; LC/BG; milestone splits
- Payment methods (NEFT/RTGS/IMPS, Cheque/DD, Gateway, Cash if legal)
- Security deposit, late interest, early payment discount

Delivery & Compliance

- Delivery date (firm/approx), location, hours; unloading responsibility
- Certificates: Phytosanitary, Fumigation, Lab Test, FSSAI, e-Way bill
- Association rulebook/agreement clauses; Force majeure

2) Accept / Negotiate / OTP Confirm

Traders can accept, counter, or request advance; accepted deals are bound via mutual OTP.

- Accept as-is → Bidirectional OTP (Trader ↔ Buyer) → Deal locked with Contract/PO.
- Counter with diffs across price/transport/packaging/payment/taxes/dates/clauses.
- Request Advance: percent/amount, reason, settlement rule.

OTP is a legal handshake; audit log captures event, IP, device, timestamp.

3) Assignment & Purpose Tagging

- Purpose: Fulfill Buyer Order / Store for Future Sale / Processing & Value Addition.
- IDs: Auto-generate or custom for Buyer/Assignment/Sub-allocation.
- Assign employees to buyers/commodities/markets; multi-region or international scope.

4) Marketplace Visit & Token Verification

Procurement employee scans Token QR/enters Token ID to view lot details.

- Lot card shows Commission Agent, Producer, Commodity/Variety/Grade.
- Bag breakdown (e.g., X=25, Z=100, W=115 bags @ 42–45 kg).
- Storage: Market stall or Storage Facility with bond number and map.
- Capture mandatory photos; Grok AI verifies visible traits and flags anomalies.
- Optional live video with Buyer/management via in-app/WhatsApp/Arattai.

5) Dynamic Bidding & Auto-Bid

Real-time bidding with stop-loss/stretch caps; AI auto-switch among buyers by rating.

```
onNewBid(bid):
    if bid.amount <= activeBuyer.maxLimit:
```

```

    autoCounter(bid.amount + increment)
else:
    alertAll(bid, tokenId)
    next = bestBuyerByRating(eligibleBuyers)
    if next and next.maxLimit >= bid.amount:
        setActiveBuyer(next)
        autoCounter(bid.amount + increment)
    else:
        awaitHumanDecision()

```

Buyer rating factors: volume, payment discipline, price discipline, dispute history, recency.

6) Sampling (Quality Verification) — Post-Bidding

- Trigger sampling when bid wins/conditionally awarded; auto or manual slot.
- Verifier check-in (timestamp/GPS), sampling method (random/systematic/stratified).
- Record moisture, impurities, broken, stems, color/ASTA, pests; attach media.
- Discrepancy → reasons (dropdown) and adjustments (price/quantity/partial rejects).

7) Weighing — Scheduled & Dual Entry

Weighing slot fixed; alerts 10 minutes prior to Agent/Trader/Operator.

- Select: Commission Agent → Token ID → Lot (e.g., 22 bags Teja Export).
- Dual entry grid: Trader vs Agent entries per bag; live mismatch detection with resolution options.
- Totals auto-calculated; session closes only via dual OTP; immutable record posted to back office (Buyer can be permissioned).

8) Finance, Taxes & Delivery Terms

- Gross = rate × net; deductions: commission/market fee/hamali/storage/sampling/adjustments.
- Taxes: IGST or CGST+SGST per jurisdiction; reverse charge; export LUT.
- Payment terms applied from deal; reminders for due dates & regulatory deadlines.

9) Allocation Across Buyers & Purposes — Example

From a 100-bag purchase from PSR & Co:

- Buyer APV (57): 22 Teja Export + 16 (334 Export) + 19 (341 C)
- Buyer SVTS (30): 30 Teja A
- Trader Inventory (13): 341 A+ for future trade

Each sub-allocation becomes its own sub-transaction with documents and ledgers, linked to the parent purchase.

10) Value Addition & Re-Tokenization

- If processed, issue child Token with hygiene/processing data, yields, re-packaging specs.
- Provenance shows Parent → Child lineage; AI recommends target markets and timing.

11) Communication & Visibility

- In-app chat/video; WhatsApp/Arattai launchers; archived links.
- Buyer access permissioned per deal (inspection, weighing, invoices, delivery docs).
- Notifications routed to assigned employees & stakeholders.

12) Back-Office & Records

- Immediate posting of: quality report, weighing summary, invoices, challans, certificates, OTP logs.
- Ledgers by Buyer/Agent/Producer/Commodity/Variety; exports (PDF/Excel); ERP/accounting APIs.

13) Alerts & SLA Timers

- Bids: cap nearing/exceeded; last 5 minutes; auction won/lost
- Sampling: slot assigned; overdue reports
- Weighing: 10-min prior; mismatch alerts; unresolved discrepancy
- Finance: payment due; regulatory filing deadlines
- Ops: overdue tasks dashboard

14) Commodity-Specific Presets (examples)

Chilli (Teja/334/341/Sannam): Bag 40–50 kg; Moisture $\leq 12\%$; Stems $\leq 3\%$; Broken $\leq 5\%$; ASTA color min; Packaging Jute/PP/Vacuum/Carton.

Turmeric (Finger/Bulb): Bag 50–75 kg; Moisture $\leq 12\%$; Curcumin %; double-lined bags.

Coriander/Cumin/Fennel/Fenugreek: Impurities %, split %, oil content (if tested); sack specs.

Pulses (Toor/Moong/Urad/Masoor): Broken %, polished/unpolished, infestation-free; PP 50 kg.

Oilseeds (Groundnut/Sesame/Soy/Castor/Sunflower): Oil %; aflatoxin limits (export); moisture $\leq 8\text{--}10\%$; jute/PP.

Cereals (Rice/Wheat/Maize): Broken %, moisture, varietal grade; gunny/PP 50 kg; FOB/delivered norms.

15) Core Data Objects (high level)

- Offer
- Deal/Contract
- Assignment
- Token & TokenChild
- Auction
- QualityReport
- WeighingSession
- Invoice/Challan/Certificates
- PaymentTerm
- Alert/Event
- Permission

16) Rules & Validations (selected)

- Price cannot exceed buyer stretch unless authorized via OTP.
- Sampling must complete before weighing (unless policy allows parallel hold).
- Weighing session locks only after dual OTP within validity window.
- Quality downgrade must map to adjustment with reason code.
- Taxes computed by jurisdiction; export LUT/duty flags respected.
- Every critical step writes an immutable audit entry.

17) Documents & Outputs

- PO/Contract
- Quality Report
- Weighing Certificate
- Tax Invoice
- Delivery Challan
- Payment Advice
- e-Way Bill (if applicable)

18) International & Multi-Region Considerations

- Currency and unit choices (kg/quintal/MT); calendars and holidays.
- Tax packs per country/state; reverse charge toggles; Incoterm-like presets.
- Localization of UI and document headers (multi-language).

Notes

This document consolidates UX, functional, and process rules for the Trader flow. It is intended as a single source of truth for design (Figma), engineering (API, data), and QA (acceptance tests).